Lab 7: Molecular Genetics — From DNA to Protein

BIOL-1

Name: ______________________ **Date:** ______________________

Overview

Every living cell follows the same basic recipe to turn genetic information into a working molecule:

> ***DNA → RNA → Protein***

This is the **Central Dogma** of molecular biology. It means your DNA is like a master cookbook that never leaves the kitchen (the nucleus). When a protein needs to be made, the cell copies just the recipe it needs into a messenger (mRNA), which travels to the ribosome — the cell's protein-building machine.

In this lab you will **be the cell**. You'll copy DNA into mRNA (*transcription*), decode mRNA into amino acids (*translation*), and discover what happens when the recipe gets a typo (*mutations*).

Learning Objectives

By the end of this lab, you will be able to:

1. Transcribe a DNA sequence into mRNA using base-pairing rules.
2. Translate an mRNA sequence into amino acids using a codon table.
3. Predict the effects of point mutations and frameshift mutations on a protein.
4. Explain the Central Dogma in your own words.

Materials

- This worksheet
- Codon table (Part 1)
- Colored pencils (optional — great for color-coding bases!)

Part 1: Base Pairing — The Language of DNA and RNA

DNA and RNA are built from four **nucleotide bases**. These bases always pair in specific, predictable ways — like puzzle pieces that only fit together one way. Understanding these pairing rules is the key to everything else in this lab.

DNA–DNA Pairing (Replication)

When a cell needs to **copy its DNA** before dividing, the double helix unzips and each strand serves as a template. The enzyme **DNA Polymerase** reads the template and adds the matching base according to these rules:

Template Base	Pairs With
A (Adenine)	**T** (Thymine)
T (Thymine)	**A** (Adenine)
G (Guanine)	**C** (Cytosine)
C (Cytosine)	**G** (Guanine)

> 🧠 ***Memory trick: A**lways with **T**, **G**oes with **C** — in DNA, A pairs with T and G pairs with C.*

Quick practice — Write the complementary DNA strand:

Template	**T**	**A**	**C**	**C**	**G**	**T**
New strand	____	____	____	____	____	____

DNA–RNA Pairing (Transcription)

When the cell needs to **make a protein**, it doesn't send the DNA out of the nucleus. Instead, the enzyme **RNA Polymerase** reads the DNA template and builds a single-stranded **mRNA** (messenger RNA) copy — a process called **transcription**.

RNA has one important difference: it uses **Uracil (U)** instead of Thymine (T).

DNA Template Base	Pairs With (RNA)
A (Adenine)	**U** (Uracil)
T (Thymine)	**A** (Adenine)
G (Guanine)	**C** (Cytosine)
C (Cytosine)	**G** (Guanine)

> 🔑 ***The only change from DNA→DNA pairing:*** *Wherever DNA has **A**, the mRNA gets **U** instead of T. Everything else stays the same.*

Quick practice — Transcribe this short DNA template into mRNA:

DNA Template	**T**	**A**	**C**	**C**	**G**	**T**
mRNA	____	____	____	____	____	____

Check your understanding:

1. What are the two base-pairing rules that are the *same* in both DNA replication and transcription?

2. What is the one key difference between DNA–DNA pairing and DNA–RNA pairing?

3. If a DNA template strand reads `G-A-T-T-C-A`, what would the mRNA sequence be?

Part 2: Transcription — DNA → mRNA

Now let's apply those pairing rules to a full gene! Transcription is like photocopying one page from a master binder — the original DNA stays safe in the nucleus while the mRNA copy travels to the ribosome.

Use the DNA→RNA pairing rules from Part 1: **A→U, T→A, G→C, C→G**.

DNA template strand: `3′ — T A C · C G T · A C G · T C G · G G T · G A C · A T T — 5′`

4. Transcribe — Write the mRNA sequence. Group into codons (groups of 3).

5. Where in a eukaryotic cell does transcription occur?

6. What enzyme carries out transcription?

Part 3: Translation — mRNA → Protein (The Codon Table)

Once the mRNA reaches the ribosome, it is read in groups of **three bases** called **codons**. Each codon specifies one amino acid (or a stop signal). This process is called **translation** — the cell is literally *translating* the language of nucleotides into the language of amino acids.

The Codon Table

> ***How to use it:*** *Find the first base in the left column, the second base in the top row, and the third base in the right column. The intersection gives you the amino acid.*

	U	**C**	**A**	**G**	
	2nd base →				*3rd base ↓*
U	**Phe** (F)	**Ser** (S)	**Tyr** (Y)	**Cys** (C)	**U**
	Phe (F)	**Ser** (S)	**Tyr** (Y)	**Cys** (C)	**C**
	Leu (L)	**Ser** (S)	⛔ STOP	⛔ STOP	**A**
	Leu (L)	**Ser** (S)	⛔ STOP	**Trp** (W)	**G**
C	**Leu** (L)	**Pro** (P)	**His** (H)	**Arg** (R)	**U**
	Leu (L)	**Pro** (P)	**His** (H)	**Arg** (R)	**C**
	Leu (L)	**Pro** (P)	**Gln** (Q)	**Arg** (R)	**A**
	Leu (L)	**Pro** (P)	**Gln** (Q)	**Arg** (R)	**G**
A	**Ile** (I)	**Thr** (T)	**Asn** (N)	**Ser** (S)	**U**
	Ile (I)	**Thr** (T)	**Asn** (N)	**Ser** (S)	**C**
	Ile (I)	**Thr** (T)	**Lys** (K)	**Arg** (R)	**A**
	🟢 **Met** (M) START	**Thr** (T)	**Lys** (K)	**Arg** (R)	**G**
G	**Val** (V)	**Ala** (A)	**Asp** (D)	**Gly** (G)	**U**
	Val (V)	**Ala** (A)	**Asp** (D)	**Gly** (G)	**C**
	Val (V)	**Ala** (A)	**Glu** (E)	**Gly** (G)	**A**
	Val (V)	**Ala** (A)	**Glu** (E)	**Gly** (G)	**G**

> ***Key:***
>
> - 🟢 ***AUG*** = *Start codon (also codes for Methionine)*

- ⛔ ***UAA, UAG, UGA*** *= Stop codons (no amino acid added — translation ends)*

Quick Practice

Try decoding these three codons before moving on:

Codon	1st base	2nd base	3rd base	Amino Acid
GCA	G	C	A	_____
UUU	U	U	U	_____
AUG	A	U	G	_____

7. Translate your mRNA from Part 2 — Use the codon table to convert each codon into an amino acid. Write the amino acid chain.

8. What was your first codon? What amino acid does it code for, and why is that codon special?

9. How did you know to stop translating? What was the stop codon?

Bonus look-up: Find two different codons that both code for Leucine (Leu). Write them here:

This is called **redundancy** — multiple codons can code for the same amino acid. Why might this be useful for an organism? *(Hint: think about mutations.)*

Part 4: Mutation Detective

Mutations are changes in the DNA sequence — like typos in a recipe. Some are harmless; others can ruin the dish. Let's investigate.

Go back to the **original DNA template strand** from Part 2.

Scenario A: Substitution (Point Mutation)

The **10th base** (the first C in the third codon ACG) changes to a **C** → **G**.

Mutated DNA template: 3′ — T A C · C G T · A G G · T C G · G G T · G A C · A T T — 5′

10. Transcribe and translate the mutated sequence.

- Mutated mRNA:
- Amino acid chain:

11. Did the amino acid chain change? Classify the mutation as *silent* (no change), *missense* (different amino acid), or *nonsense* (creates a premature stop).

Scenario B: Insertion (Frameshift Mutation)

Return to the **original DNA template**. An extra **G** is inserted right after the 3rd base (after the C in TAC).

Mutated DNA template: 3′ — T A C · G C G · T A C · G T C · G G G · T G A · C A T · T — 5′

12. Transcribe and translate the frameshift-mutated sequence.

- Mutated mRNA:
- Amino acid chain:

13. Compare the frameshift protein to your original. How many amino acids changed? Why is this type called a "frameshift"?

Scenario C: Deletion (Another Frameshift)

Return to the **original DNA template**. This time, **delete the 7th base** (the `T` in the second codon `CGT`).

Mutated DNA template: `3′ — T A C · C G A · C G · T C G · G G T · G A C · A T T — 5′`

(Re-group into codons after the deletion:) `3′ — T A C · C G A · C G T · C G G · G T G · A C A · T T — 5′`

14. Transcribe and translate the deletion-mutated sequence.

- Mutated mRNA:
- Amino acid chain:

15. How does a deletion compare to an insertion? Did both cause the same amount of damage to the protein?

Comparing All Three Mutations

16. Rank the three mutations (substitution, insertion, deletion) from least to most damaging. Explain your reasoning.

Part 5: Big Picture

17. Complete the Central Dogma:

DNA → ___ → ___

18. Where in a eukaryotic cell does transcription occur?

19. Where does translation occur?

20. What cellular structure reads mRNA and builds proteins (carries out translation)?

21. In your own words: How can a single change in someone's DNA affect their physical traits? Trace the path from gene → protein → trait.

Part 6: Real-World Connection

22. Sickle-cell disease is caused by a single point mutation in the hemoglobin gene. The 6th amino acid changes from Glutamic acid (Glu) to Valine (Val). Based on what you learned today:

- What kind of mutation is this — silent, missense, or nonsense?
- Why can a single amino acid change affect the shape of an entire protein? *(Hint: amino acids fold into 3D shapes, and each one contributes to the fold.)*

23. Not all mutations are bad! Can you think of a situation where a mutation might actually help an organism survive? *(Hint: think about how evolution works.)*

Bonus Challenge 🧬

Design your own mutation! Start with the original DNA template from Part 2. Create a mutation of your choice (substitution, insertion, or deletion), then transcribe and translate it.

- Type of mutation you chose:
- Your mutated DNA template:
- Mutated mRNA:
- Amino acid chain:
- What happened to the protein? Was it silent, missense, nonsense, or a frameshift?